Standardization & Normalization

- when dealing with numerical features we have different Ranges for them:

Like age is 1 - 100 ish

- BMI ~18 < < 30ish.

So our model Don't Know about the Content So we need to make them, make sense in a Suitable Range. Consistence Scale.

transform.

Standard Scaler & MinMax Scaler

→ Choose one of them.

- fit_transform → train set
- transform → on test set.

Scalling should only happen on numerical features.

for each column.

mean ~ 0 , std. dev. ~ 1

N(0, 1) → Standard Scaler

if you are working with models that are based on Gradient Descent using standard scaler is great way to Reach Global optimals faster

$$Z = \frac{X - \bar{X}}{\sigma}$$

۱۳ فروردین پنجشنبه ۱۳ شوال ۱۴۴۷ | APRIL Thursday 2026 | 2

normalization. mini Value = 0

$$x\_scaled = \frac{x - x.min}{x.max - x.min}$$

uniformly distributed.

the max became 1, the min become 0!

we should save the scalers, ~~~
into file

روز طبیعت (تعطیل)

۱۴ فروردین جمعه ۱۴ شوال ۱۴۴۷ | APRIL Friday 2026 | 3

mo	tu	we	th	fr	sa	su	mo	tu	we	th	fr	sa	su
		1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24	25	26
27	28	29	30										